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COMPUTER NETWORKS LAB

Lab Report

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EXPERIMENT 1

INTRODUCTION TO PACKET TRACER, PEER-TO-PEER COMMUNICATION, STUDY OF CABLES AND ITS COLOR CODES

Date: 23-10-2026

AIM:

To learn about Cisco Packet Tracer, connect two PCs using a switch and check whether they can communicate using ping, and to study straight-through and crossover cables with their color codes.

SOFTWARE USED:

Cisco Packet Tracer

THEORY:

Cisco Packet Tracer is a network simulator by Cisco. With it we can build networks, configure devices and test them without using real hardware. In peer-to-peer communication, the computers talk to each other directly and no server is needed. A switch connects the devices inside one network and sends the data to the correct device using the MAC address.

An Ethernet cable has 8 wires (4 twisted pairs) and RJ-45 connectors are fixed at both ends. There are two wiring standards, T568A and T568B.

T568A: White-Green, Green, White-Orange, Blue, White-Blue, Orange, White-Brown, Brown

T568B: White-Orange, Orange, White-Green, Blue, White-Blue, Green, White-Brown, Brown

Straight-through cable – the same standard is used on both ends (T568B – T568B). It connects different devices, for example PC to switch or switch to router.

Crossover cable – T568A on one end and T568B on the other end. It connects similar devices, for example PC to PC, switch to switch or router to router.

IP ADDRESSES USED:

PC0 – 192.168.1.1, Subnet mask 255.255.255.0

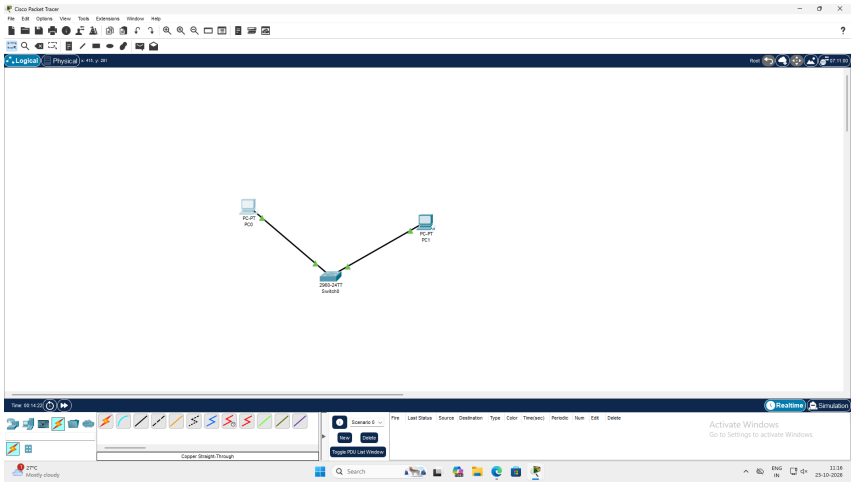
PC1 – 192.168.1.2, Subnet mask 255.255.255.0

PROCEDURE:

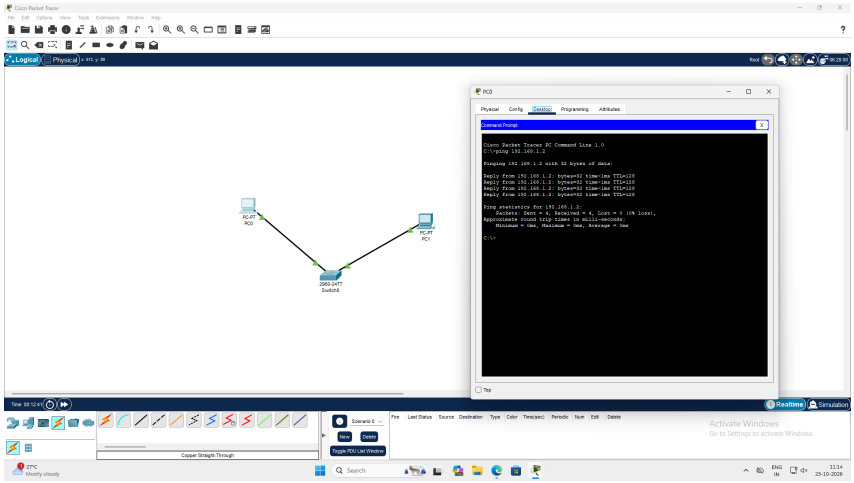
1. Open Cisco Packet Tracer.
2. Take two PCs (PC0 and PC1) from End Devices and one 2960 switch from Switches and place them in the workspace.
3. Select the Copper Straight-Through cable and connect PC0 to Fa0/1 of the switch and PC1 to Fa0/2 of the switch.
4. Click on PC0, go to Desktop, then IP Configuration and enter IP address 192.168.1.1 and subnet mask 255.255.255.0.
5. Click on PC1, go to Desktop, then IP Configuration and enter IP address 192.168.1.2 and subnet mask 255.255.255.0.
6. Wait until all the link lights turn green.
7. Open PC0, go to Desktop, then Command Prompt and type ping 192.168.1.2
8. Check the replies.

OUTPUT:

Topology:



Ping from PC0 to PC1:



RESULT:

Thus the two PCs were connected using a switch in Cisco Packet Tracer and the ping from PC0 to PC1 was successful (4 sent, 4 received, 0% loss). The straight-through and crossover cables and their color codes were studied.

EXPERIMENT 2

ROUTER CONFIGURATION (CREATING PASSWORDS, CONFIGURING INTERFACES)

Date: 23-10-2026

AIM:

To configure a router by setting passwords and giving IP addresses to its interfaces, and to check the communication between two PCs in different networks through the router.

SOFTWARE USED:

Cisco Packet Tracer

THEORY:

A router is used to connect two or more different networks. It forwards the packets based on the IP address. Each interface of the router belongs to a different network and works as the default gateway for the PCs in that network.

Passwords are set on the router for security. The console password is for console access, the enable password is for privileged mode and the VTY password is for remote (Telnet) access. By default the router interfaces are shut down, so we have to use the `no shutdown` command to turn them on.

IP ADDRESSES USED:

Router Gig0/0 – 192.168.1.1 / 255.255.255.0

Router Gig0/1 – 192.168.2.1 / 255.255.255.0

PC2 – 192.168.1.2 / 255.255.255.0, Gateway 192.168.1.1

PC3 – 192.168.2.2 / 255.255.255.0, Gateway 192.168.2.1

PROCEDURE:

1. Open Cisco Packet Tracer and place one 2911 router and two PCs (PC2 and PC3).
2. Connect PC2 to Gig0/0 and PC3 to Gig0/1 of the router using copper cables.
3. Click on the router and go to the CLI tab. Type no for the initial configuration dialog and press Enter.
4. Type the following commands to set the passwords:

```
Router>enable
Router#configure terminal
Router(config)#hostname R1
R1(config)#enable password cisco
R1(config)#line console 0
R1(config-line)#password cisco
R1(config-line)#login
R1(config-line)#exit
R1(config)#line vty 0 4
R1(config-line)#password cisco
R1(config-line)#login
R1(config-line)#exit
```

5. Type the following commands to configure the interfaces:

```
R1(config)#interface gig0/0
R1(config-if)#ip address 192.168.1.1 255.255.255.0
R1(config-if)#no shutdown
R1(config-if)#exit
R1(config)#interface gig0/1
R1(config-if)#ip address 192.168.2.1 255.255.255.0
R1(config-if)#no shutdown
R1(config-if)#end
R1#show ip interface brief
```

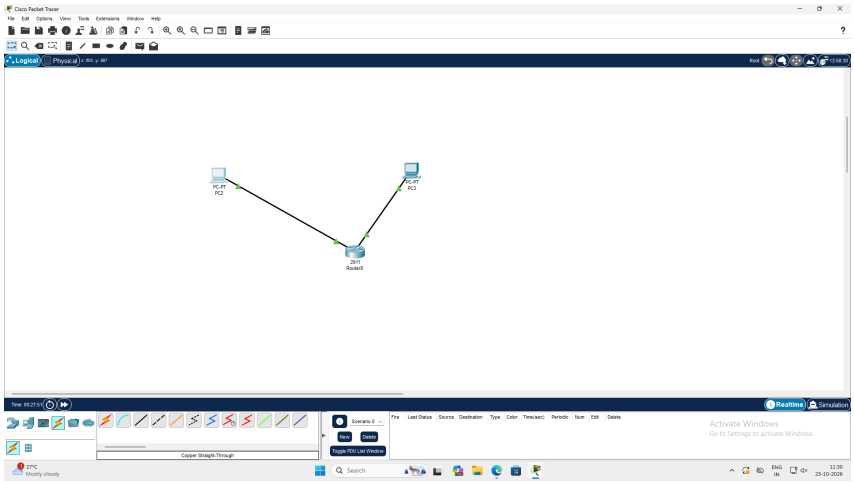
6. In PC2 give IP 192.168.1.2, subnet mask 255.255.255.0 and default gateway 192.168.1.1.

7. In PC3 give IP 192.168.2.2, subnet mask 255.255.255.0 and default gateway 192.168.2.1.

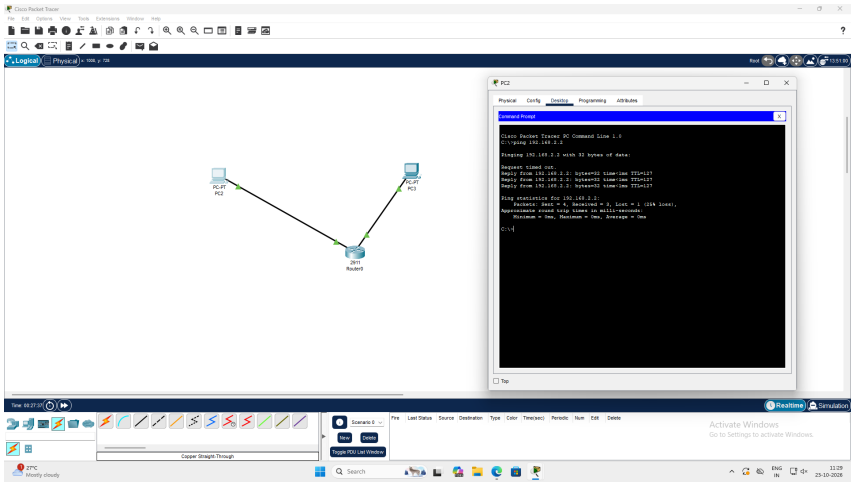
8. Open Command Prompt in PC2 and type ping 192.168.2.2

OUTPUT:

Topology:



Ping from PC2 to PC3:



The first request timed out because of ARP, after that all replies came.

RESULT:

Thus the router was configured with passwords and IP addresses on its interfaces, and the two PCs in different networks were able to communicate through the router.